

I. Decision Tree Fundamentals (1–10)

1. **Q:** What is the fundamental goal of a Decision Tree algorithm in supervised learning?

A: To **partition the feature space** into regions where the output variable can be predicted effectively.

2. **Q:** What three key components make up the flow chart-like structure of a decision tree?

A: **Root nodes, decision/split nodes, and leaf nodes.**

3. **Q:** In the feature space, what is the key objective of a decision tree?

A: To **split the space into regions** where the target variable is as homogeneous as possible.

4. **Q:** What are the final domains in the feature space partitioning called?

A: **Regions or leaves.**

5. **Q:** Decision trees are versatile because they can be used for which two main types of problems?

A: **Regression and classification.**

6. **Q:** What does CART stand for?

A: **Classification and Regression Trees.**

7. **Q:** For regression, what type of value is assigned to a datapoint in a leaf?

A: A **constant numeric value**, typically the **mean** of the responses in that leaf.

8. **Q:** If the feature space is partitioned into (M) regions $(R_1, \dots, R_M)$, what is the regression prediction $\hat{y}$ for $\vec{x}$?

A:

$$\begin{aligned} &[\\ \hat{y} &= \sum_{k=1}^M c_k I(\vec{x} \in R_k) \\ &] \end{aligned}$$

9. **Q:** In the formula $\hat{y} = \sum_{k=1}^M c_k I(\vec{x} \in R_k)$, what does $I(\vec{x} \in R_k)$ represent?

A: An **indicator function**: 1 if $\vec{x} \in R_k$, 0 otherwise.

10. **Q:** What does (M) represent?

A: The **number of regions (or leaves)** in the decision tree.

II. Optimal Tree Construction (11–20)

11. Q: To obtain optimal (c_k), which loss function must be minimized?

A: **Sum of Squared Errors (SSE).**

12. Q: Formula for SSE for (n) datapoints and (M) regions?

A:

$$\text{SSE}(\vec{c}) = \sum_{k=1}^M \sum_{\vec{x}_i \in R_k} (y_i - c_k)^2$$

13. Q: Setting $(\frac{\partial \text{SSE}}{\partial c_k} = 0)$, what statistical property does (c_k) become?

A: The **mean of the target values** in region (R_k).

14. Q: Equation for optimal constant ($\hat{c}_k$) in region (R_k)?

A:

$$\hat{c}_k = \frac{1}{|R_k|} \sum_{\vec{x}_i \in R_k} y_i$$

15. Q: What does this average represent?

A: The **predicted response** for all points in that region.

16. Q: What does "greedy" imply in tree construction?

A: The algorithm **chooses the best split at each step** without considering future splits.

17. Q: Goal of the algorithm when deciding splits?

A: To **minimize SSE** within the resulting child regions.

18. Q: Three main steps to obtain optimal (c_k)?

A: **Split the space, calculate optimal (c_k) for each region, assign predictions.**

19. Q: Two key steps performed iteratively in greedy algorithm?

A: **Find best split** (feature and split point) and **partition the region.**

20. Q: In the material selection example, what determines the path from root to leaf?

A: **Feature values and decision rules** at each node.

III. Mathematical and Practical Aspects (21–30)

21. Q: In $(\sum_{i=1}^n y_i I(\vec{x}_i \in R_k) = \hat{c}_k \sum_{i=1}^n I(\vec{x}_i \in R_k))$, what does $(\sum_{i=1}^n I(\vec{x}_i \in R_k))$ represent?

A: The **number of datapoints in region (R_k).**

22. Q: Term (y_i) represents what in SSE for regression?

A: The **true response value** for datapoint (i) .

23. Q: If region (R_k) is empty, what happens to $(\hat{c}_k)$?

A: It **cannot be computed** (division by zero).

24. Q: What is the central concept minimized to achieve homogeneity?

A: **Sum of Squared Errors (SSE)** or **variance within regions**.

25. Q: For classification trees, which metric is minimized instead of SSE?

A: **Gini index** or **entropy**.

26. Q: In material selection example, decision nodes ask about which three properties?

A: **Material type, operating temperature, and strength**.

27. Q: What does "piecewise constant" mean?

A: The **prediction is constant within each leaf/region**, changes only between regions.

28. Q: After a split, what is the criterion for selecting optimal split point and feature?

A: The split that **minimizes the combined SSE** of the resulting child regions.

29. Q: To prevent overfitting, what process is applied?

A: **Pruning** or **setting a maximum depth / minimum samples per leaf**.

30. Q: Which type of node holds the final prediction value (c_k) ?

A: **Leaf node**.